

Assessment of the public acceptance and utilization of renewable energy in Pakistan

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ABSTRACT

The acceptance of renewable energy technologies is a complicated and multifaceted process influenced by a broad range of factors. Therefore, this study aims to examine the factors influencing consumer intention to utilize renewable energy (RE). Moreover, the current research highlights the factors that encourage or discourage consumers from utilizing RE by expanding the structural context of the Theory of Planned Behavior (TPB) by integrating three new considerations (the perception of self-effectiveness, beliefs about the benefits of RE, and perception about neighbor participation). The data used for analysis were collected from 351 households in four large cities, including Rawalpindi, Lahore, Gujranwala, and Faisalabad, in Pakistan. We utilized the Structural Equation Modeling (SEM) approach to check the relationship between constructs and latent variables. The results reveal that the driving factors, i.e., the perception of self-effectiveness, awareness, and perception about neighbor participation have significant and positive effects on consumer intention to utilize RE. However, consumer beliefs related to the cost of RE utilization have a negative effect on their intention to utilize RE. More interestingly, it was observed that beliefs about the benefits of RE and environmental concern have insignificant effects. The outcomes of this study can assist policy makers, experts and consumers in understanding renewable energy consumption and gaining awareness about environmental problems while simultaneously improving environmental sustainability practices.

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1. Introduction

Due to the recent economic growth and increasing population, energy demand has increased tremendously. Many developing countries, including Pakistan, still rely on fossil fuels for energy generation. Nonetheless, due to their high prices and adverse environmental effects, fossil fuels can no longer be depended on as a primary energy source (Iqbal et al., 2018). Governments are seeking economical and sustainable sources to resolve the existing energy crisis and monitor the current economic and environmen-

tal situation (Merino et al., 2020; Ikram et al., 2019). The change from traditional forms of energy generation to sustainable sources is due to the need for RE production and consumption of the public sector and regulatory bodies (Liu et al. 2013). Environmental degradation and climate vulnerability have led policy makers to realize other possible methods of energy production to minimize fossil-fuel-based carbon footprints (Azapagic and Stichnothe, 2011).

Various countries have taken initiatives to increase the share of RE in their overall energy composition (Ikram et al., 2020). RE is considered a true alternative to traditional energy sources due to its decreasing costs throughout the world (Irfan et al., 2020a). RE programs are believed to enhance residents' quality of life in various ways. First, RE sources are likely to reduce emissions from air pollution and greenhouse gas (GHG) (Rehman et al., 2020; Irfan et al., 2019a), thus improving people's health. Second, RE

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penetration creates new jobs in the local population and directly promotes the domestic economy (Ram et al., 2020). Some RE sources are slightly costly, as in the case of photovoltaics, in which the cost of such energy generation is higher than that of other RE and non-RE sources (hydropower and coal). PV panels lose efficiency due to dust and pollution. The regular cleaning of PV panels, in turn, demands substantial costs (Sharma et al., 2012). However, due to resource scarcity and the increasing rate of fossil fuel consumption, thermal energy (generated from coal, oil, and natural gas) is becoming more expensive (Käberger, 2018). Shah et al. (2018) assessed the economic feasibility of off-grid PV solutions for the rural regions of Pakistan and found that the energy generated by off-grid PV systems only costs Pakistani rupees (PKR) 7.98/kWh, which is much more economical than traditional energy, which costs PKR 20.79/kWh. Finally, due to technological advancements and the potential for cost reductions, RE is outcompeting non-RE, providing affordable energy and stabilizing energy costs in the future (Kaygusuz, 2012; Caspary, 2009).

RE meets the growing population's energy requirements in a sustainable way (Irfan et al., 2019b). RE development could be expanded by understanding consumers' behavior towards RE. Several studies have examined the intention of consumers to use RE in developed countries. The results of their studies revealed mild to strong public acceptance of RE technologies (Hamilton et al., 2018; McGowan and Sauter, 2005). Some scholars have concentrated on the structural capacities behind the public acceptance of RE because the key determinants are economic motivation, efficient policies, and support programs (Paravantis et al., 2018; Musall and Kuik, 2011). Other groups of researchers have examined psychologically oriented consumer behavior (Huijts et al., 2012) and utilized quantitative assessments and environmental activism in large measures (Kaufmann et al., 2012). The intention to use RE is measured as the perception of residents' attitudes (Zografakis et al., 2010), specifying that people's attitudes determine the degree to which RE is used. The contributing factors that influence public acceptance of RE have been investigated in successive studies. Wüstenhagen et al. (2007) found that consumer intention rises with positive behavior toward RE and that it will lessen environmental issues and decrease as the cost of energy increases. These findings support the study conducted by Irfan et al. (2020)b.

Similarly, family size, income (Hansla et al., 2008), social status (Batley et al., 2001), awareness, personal experience, and education (Batley et al., 2000) are also important factors. Other factors include location and urban vs. rural users. Devine-Wright (2005) reported that location is a significant factor that influences the social acceptance of wind farms. Shakeel and Rahman (2018) found that the social acceptance rate of RE is very high among urban residents in Pakistan. Liu et al. (2013) conducted a survey in the rural areas of China and found that rural dwellers are willing to pay for RE.

Several scholars have contributed to the Theory of Planned Behavior (TPB), as some of them have focused on "environmental concern" to determine consumers' knowledge about environmental problems and their motivation to solve these problems (Čábelková et al., 2020; Kim and Choi, 2005), while other groups of researchers have focused on the "cost of RE" and how it affects consumers' decisions (Shakeel and Rahman, 2018; Klepacka, 2019). Similarly, some scholars have focused on the "awareness of RE" (Sarin et al., 2018) and "attitude towards RE" (Dalton et al., 2008). Despite the long-standing interest of researchers, they largely overlooked to examine the novel factors (that this research has introduced), which might shape public behavior. Consequently, there is a clear room to identify and analyze these factors, influencing consumers' intention to utilize RE. Contrary to prior studies, the current research has anticipated all these factors from the viewpoint of considering RE. In this way, this research has contributed

to the existing knowledge base by bridging all these research gaps.

The contributions of this study are threefold. First, this study fills the research gap by finding the domestic energy provision and the possible factors encouraging or restricting Pakistani households from using RE. Second, the behavioral framework of the theory of planned behavior has been expanded by incorporating three additional factors: perception of self-effectiveness, beliefs about the benefits of RE, and perception about neighbor's participation. Third, the current study has extended the research results differently than the prior studies. For example, as mentioned before, these novel factors have never been considered as possible influencing factors of RE utilization. Along these lines, the perception of self-effectiveness is shown to be an important driver in the way of utilizing RE. Similarly, perceptions about neighbor participation also persist as a significant addition to the behavioral framework of the theory of planned behavior. In contrast, beliefs about the benefits of RE have been found to play an insignificant role in RE utilization. We intend to focus on the general character of RE technologies in this study, as it has a positive environmental impact but demands a higher participation cost compared with conventional sources of energy.

Finally, concerning the relevance of the results, though the findings are based on a survey conducted in a specific geographical location (Pakistan), the implications of these results indicate the phenomenon of consumers' intention-based factors to accept and utilize RE. Pakistan has an abundance of RE sources in the form of solar, wind, biomass, geothermal, and hydroelectric power. With such abundant RE sources, if harnessed effectively, the country can meet 100% of its current and future energy demand. However, Pakistan is a developing country and not particularly known for its support of RE due to several challenges and barriers, including economic, policy, technological, and social barriers (Irfan et al., 2019b). The country primarily depends on fossil fuels for energy generation. The national economy has been heavily burdened by imported costly fossil fuels to generate energy. Pakistan's weak economy cannot withstand such a burden. Consequently, the country is experiencing severe energy crisis (Solangi, et al., 2019). In addition, the society in Pakistan is integrated, stipulating that the acts of peers, family, and neighbors have a significant influence on the beliefs of local dwellers (Shakeel and Rahman, 2018). Therefore, people have concerns regarding self-effectiveness, neighbor participation, and the benefits of RE. Being a developing country, it is anticipated that these novel factors may also exist in other developing countries, along with the prevailing energy crisis. In this way, the research results obtained in this case study are not case-specific but rather are transferrable to other countries and emphasize the importance of the effect of accompanying factors on RE utilization. Consequently, Pakistan would be considered as a representative model for other countries to recognize this phenomenon. Furthermore, the lessons learned from this study will help other developing countries and emerging economies when devising strategies and policies regarding RE deployment in their respective countries.

The rest of the paper is organized as follows. Section 2 presents the literature review. The methods, research design and hypothesis formulation are reported in Section 3. Section 4 compiles the results and discusses the research findings. Finally, the concluding remarks, policy implications and limitations of the study are presented in the last section of the paper.

2. Literature review

Recently, some scholars have analyzed consumer intention to use RE technologies in developing countries. Hosseini et al. (2018) compared consumer intention to use RE technologies in different countries by employing the social acceptance pyramid.

The research findings reveal that wind energy has a higher level of social acceptance than do geothermal and solar energy technologies. Alam et al. (2014) scrutinized consumer intention to utilize RE in Malaysia and found that several factors, such as perceived behavioral control, simplicity of use, relative advantage, price reduction, and awareness, significantly influence the intention to use RE.

Vand et al. (2019) conducted a survey to investigate the attitude of consumers in supporting RE in China. The outcome of the study indicated that low awareness and income level are major barriers to RE utilization. Moreover, 97% of respondents showed a willingness to completely or partly alter their sources of energy generation after being made aware of the negative consequences of fossil-fuel-based energy. Majhi and Banarjee (2019) conducted research in India to determine consumer acceptance of the transition to RE utilization. The research findings revealed that consumers demonstrate significant interest in RE adoption and are worried about the environmental threats brought about by fossil fuels. Düşteğör et al. (2015) conducted a study in Saudi Arabia and observed that the major influencing factors of RE adoption are financial incentives and government subsidy programs.

There have been few studies directed in Pakistan to examine the intention of residents to accept and utilize renewable energy generation technologies (RETs). Irfan et al. (2019c) highlighted that Pakistan is struggling with severe energy crisis and that off-grid solar power is the best possible solution for overcoming the prevailing energy crisis in the country. Wang et al. (2020) assessed the various RE sources in the country to diversify its energy generation from fossil fuels to that from RE. The above authors found that there is enormous potential for wind, solar, and biomass energy in the country. Akhtar et al. (2018) conducted a study to assess the supply and future growth of various RE sources and uncovered that Pakistan has the potential to generate 150,000 MW of wind energy, 56,000 MW of hydroelectricity, and 50,000 MW of solar energy respectively.

Raza et al. (2020) proposed policy guidelines for developing RETs in Pakistan's rural regions. Mirza (2015) documented residents' perceptions and satisfaction with different RETs. The research results specify that local dwellers have a high acceptance of solar energy. Abdullah et al. (2017) conducted a survey to determine the public willingness to accept a solar home system (SHS), the outcomes of which indicated that 81% of participants exhibited a high willingness to pay for SHSs. However, there are multiple barriers that hinder residents from using SHSs, such as low awareness, risk, and the high price of solar panels.

This literature review helps in understanding consumers' awareness and their intention regarding RE as a sustainable energy source to mitigate climate change. Next, we review the research framework and formulate hypotheses leading up to the analysis and presentation of the results in the subsequent sections.

3. Methods

3.1. Modeling framework

Prior studies have indicated that customers' decision-making process is based on a dynamic series of subprocesses. Scholars have suggested different conceptual models, for instance, self-efficacy theory, reasoned action theory, and the TPB, to analyze the complicated essence of buyers' purchase intention (Procter et al., 2019; Parkinson et al., 2017; Liu et al., 2013). However, the TPB has been most widely employed by researchers in many consumer research domains, for example, e-commerce, organizational design, and green behavior, to describe and forecast consumer behavior (Si et al., 2020). Researchers have identified that compared with other models, the TPB successfully examines consumer behavior in

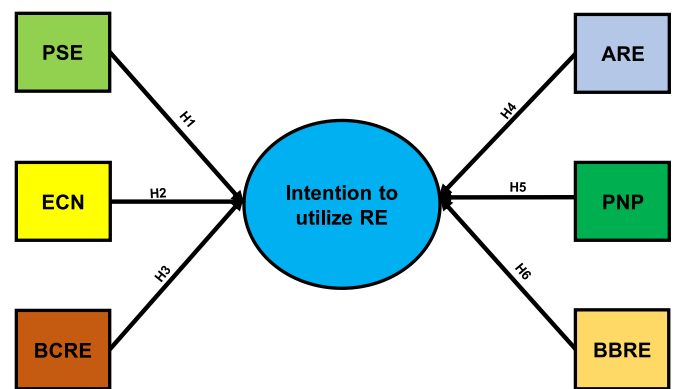


Fig. 1. Modeling framework indicating the influencing factors of consumer intention to utilize RE. **Notes:** PSE: Perception of self-effectiveness; ARE: Awareness; ECN: Environmental concern; PNP: Perception about neighbor participation; BCRE: Beliefs about the cost of RE utilization; BBRE: Beliefs about the benefits of RE.

different settings. As, we have focused on consumer behavior in this research, therefore, the TPB has been utilized to establish the modeling framework.

According to the TPB, the behavioral intentions of an individual regulate his/her behavior. Before exhibiting a certain behavior, individuals first weigh their implications and then execute the behavior, which leads them to their desired outcome. The striking beliefs of an individual and the forecasts of the outcomes associated with a given behavior form his/her attitude (Ajzen, 1991). In terms of economic, political, and social characteristics, scholars believe that several elements affect the approval of a specific technology. However, the TPB only emphasizes cultural, behavioral, and managerial views to assess consumer intention to partake in a certain behavior and excludes certain considerations that may affect their intentions (Sniehotta et al., 2014; McEachan et al., 2011). As the awareness of RE in Pakistan is in the developing stage, residents are worried about social standards, the environment and the benefits of using RETs.

Consequently, we have expanded the TPB by including three integral factors such as the perception of self-effectiveness, beliefs about the benefits of RE, and perception about neighbor participation. The perception of self-effectiveness is defined as “one’s ability to effectively perform a certain action/behavior, and this particular action/behavior will be appreciated by society”. The intention and behavior of a person are directly and indirectly affected by the perception of self-effectiveness. Similarly, beliefs about the benefits of RE are defined as “an individual’s evaluation of the positive outcomes of consuming RE in the form of climate mitigation, reducing air pollution and improving the health conditions of people”. The perception of neighbor participation refers to “consumers’ intentions and how they are shaped by what a neighbor does”.

We have assessed consumer behavior by assessing consumer intention to participate in the utilization of RE. The modeling framework of this study is presented in Fig. 1. The analytical framework in accordance with existing and new variables will comprehensively assist in examining consumer intention to utilize RE.

3.2. Formulation of hypotheses

3.2.1. Perception of self-effectiveness

The pertaining actions will not exist if individuals are unable to exhibit a certain behavior (Irfan et al., 2020c). The perception of self-effectiveness (PSE) directly influences consumer intention and indirectly influences consumer behavior. In view of consumer intention to use RE, the PSE is related to the ease or difficulty

perceived by consumers in using RE. One major challenge that may affect the PSE of individuals is the struggle linked to technology acceptance. For such acceptance, installation, and utilization of RE, the key resources needed should be available to consumers (Yun and Lee, 2015). Previous studies have shown that consumer intention to accept RETs is positively shaped by their PSE. For instance, Wojuola and Alant (2017) conducted a study in Nigeria and investigated whether the PSE is the main driver behind the acceptance of RETs. Hegan and Pijawka (2015) studied public perceptions regarding climate vulnerability and support for RE in North America. The research results specified that people are worried about the environmental problems caused by the burning of fossil fuels and have a great deal of support for RE. Moreover, their support for RE is significantly influenced by their PSE. Similarly, consumer intention regarding the intake of biofuels has been examined by Zhang et al. (2011), and the results show that the PSE is a critical factor in consumers' decisions regarding biofuel use. By considering these research findings, the first hypothesis is formulated as follows:

H1: There is a positive relationship between the PSE and consumer intention to utilize RE.

3.2.2. Environmental concern

The degree to which citizens are aware of environmental issues and are interested in solving them is known as environmental concern (ECN) (Ndebele, 2020). Globally, people are conscious of their regular practices of consumption and of the level at which their consumption practices impact the environment (Yue et al., 2020). Individuals who exhibit such behavior often display a positive approach to RETs and attempt to safeguard the environment (Kalkbrenner and Roosen, 2016). ECN and its impact on the choice of buyers to purchase green products (solar water heaters, solar lights, solar cookers, solar tube wells, biomass stoves, and biomass boilers) have recently received increasing attention from researchers. Environmental concerns are less important than the desire to use less energy to lower the costs for households of buying energy (Shakeel and Rahman, 2018). Moreover, neither concerns about the environment nor the desire to pay less for energy may be of concern to households that do not have access to energy or that suffer from the chronic disruption of the energy supply. Such households may be inclined to utilize RE sources if they can afford to Irfan et al. (2020)b. Conversely, some researchers consider ECN a significant element that affects consumer intention to accept and use RE. Consumers who possess positive ECN track the use of energy carefully and have an optimistic attitude toward RETs (Karlström and Ryghaug, 2014). They consider resource management a self-declared responsibility (Chen and Knight, 2013). Wang et al. (2018) claimed that consumers' knowledge of climate vulnerabilities could affect their decision to use RE. A further study reports the impact of ECN on consumer intention to preserve energy (Hartmann and Apaolaza-Ibáñez, 2012). Considering these results, the choice of buyers to use RE is likely to be positively influenced by ECN and devised as follows:

H2: There is a positive relationship between ECN and consumer intention to utilize RE.

3.2.3. Beliefs about the cost of renewable energy

Lee and Heo (2019) reported that individuals take into account cost information in assessing the losses linked to the purchasing procedure. Saving on energy costs and subsidies for the purchase of equipment that utilizes RE are the primary motives involved in utilizing some RE sources (Klepcka, 2019; Klepcka et al., 2018). High cost is the largest barrier to RE acceptance, as reported by Iskin et al. (2013). Although the cost of RETs has decreased considerably over recent years (Mohandes et al., 2019), they are still more expensive than fossil-fuel-based conventional

energy. RE ventures require massive costs as they escalate investments (Irfan et al., 2019d). Here, cost refers to both monetary and nonmonetary costs. The results of various studies imply that a negative connection exists between cost and consumer intention to purchase RE. Sardanou and Genoudi (2013) assessed the determinants of consumers' willingness to adopt RETs, and it was found that cost is the main obstacle to RE adoption. Kostakis and Sardanou (2012) revealed that consumer intention decreases as the cost of RETs increases. Moula et al. (2017) also exposed that high costs constitute a major barrier to RE adoption. Capital cost has recently been shown to be the main barrier to RE acceptance (Komendantova and Yazdanpanah, 2017). Based on these arguments, the third hypothesis is devised as follows:

H3: There is a positive relationship between BCRE and consumer intention to utilize RE.

3.2.4. Awareness

Awareness is an important element in residents' minds when adopting new technology (Jabeen et al., 2019; Gordon, 2010). From the RE perspective, awareness is regarded as consumers' knowledge about RE in terms of cost, capability, and other related issues (Khambalkar et al., 2010). Awareness plays an important role in the process of consumer decision making regarding whether to accept and utilize RE (Ali et al., 2019). Al-Marri et al. (2018) led a study in Qatar and found that consumer intention to consume RE is positively linked with awareness. Abdullah et al. (2017) conducted research in Pakistan and reported that consumers who are conscious about RE contributing to carbon mitigation may take significant actions (shift their routines from traditional room/water heating to RE-based room/water heating, i.e., solar room/water heaters, and move from conventional ways of electrifying the house to solar home systems) to promote an environment that is free of pollution. However, consumers often remain unaware of significant environmental and energy conservation concerns (Lopes et al., 2019). Therefore, this lack of awareness is a major reason for the adverse impact on consumer intention to use RE. The fourth hypothesis is formulated on the basis of these claims:

H4: There is a positive relationship between ARE and consumer intention to utilize RE.

3.2.5. Perception about neighbor participation

Studies have shown that social norms influence RE acceptance. The societal pressure to act in a specific manner is always perceived (Steg et al., 2015). Perception about neighbor participation (PNP) is regarded as the influence on RE use by neighbors and has been reported as the main influencer of consumers' energy conservation intentions. Opiyo et al. (2019) observed a positive association between perceptions about neighbor participation and residential attitudes toward SHSs. Kosugi et al. (2019) investigated the influence of the neighborhood on the diffusion of residential PV systems in Japan and found that neighbors have a significant effect on consumers' beliefs and perceptions, which shape their decisions regarding the adoption of PV systems. Palm (2017) examined the influence of peers on the acceptance of small PV systems among Swedish users. The outcome of their study uncovered that the opinions of neighbors, friends, and family are important to consumers when adopting PV systems for household purposes. Rai et al. (2016) found positive links between neighbor involvement and consumer intention regarding the preservation of energy and the environment. PNP has been unveiled by Qureshi et al. (2017) as an important motivating factor for residents in accepting RETs. It is expected that the same can be true in this research and is hypothesized as follows:

H5: There is a positive relationship between PNP and consumer intention to utilize RE.

3.2.6. Beliefs about the benefits of RE

Consumers' understanding of the advantages offered by RETs usage in the form of carbon mitigation, the security of resources, and energy efficiency is referred to as their beliefs about the benefits of RE (BBRE) (Süsser and Kannen, 2017). Consumers compare the benefits of RETs with those of conventional sources of energy and take actions considering socioeconomic perspectives (Aitken, 2010). Consequently, efforts should be made to increase the knowledge of consumers about the advantages associated with RE consumption (improvement in air quality and reduction in carbon footprints) and inform them about the adverse impact of thermal energy consumption (Islam, 2014; Mohsin et al., 2019). Stigka et al. (2014) examined the social acceptance of RE sources as an alternate solution to fossil-fuel-based energy generation and found that consumers place importance on the use of RE sources due to the perceived benefits and their contribution in achieving sustainability. Finally, the sixth hypothesis has been devised considering these findings as follows:

H6: There is a positive relationship between BBRE and consumer intention to utilize RE.

3.3. Research design

3.3.1. Sample size, characteristics, and selection criteria

This study covers the four large cities of Pakistan—Rawalpindi, Lahore, Gujranwala, and Faisalabad—due to their high population density and modern facilities. In addition, they exhibit the country's distinctive economic growth characteristics. A well-structured survey was administered in these cities during May, June, and July of 2019. A detailed description of the questionnaire is provided in Appendix A1. The questionnaire was tested by conducting a pilot survey on a smaller sample size before the start of the actual survey to obtain accuracy and generate more meaningful results (Awadh et al., 2014). An in-depth explanation was provided to participants of each element of the questionnaire in this context. The authors approached the respondents in-person to conduct the survey. Later, a convenient random sampling technique was used to select respondents (Etikan et al., 2016) to ensure that all members of the population have an equal opportunity to be selected for the survey. A total of 400 questionnaires were delivered, and 351 valid responses were received, showing a response rate of 87.70%. Incomplete questionnaires were not entertained in the data analysis and were considered invalid. The composition of the sample in the context of the total and urban population of Pakistan is 351:221,483,664¹:77,437,729, respectively. The sample presents a balanced proportion from each sector of life, as respondents were selected from all age, income, education and occupation categories. Osborne and Costello (2004) found that no specific criterion has been followed for deciding on sample size in behavioral studies, as one-sixth of the studies applied a 2:1 subject-to-item ratio, and 20% of the studies applied less than a 5:1 subject-to-item ratio. Our sample size (351 respondents) is even more than a 5:1 subject-to-item ratio, which is sufficient for performing the data analysis. The questionnaire comprises two sections of questions: (i) questions regarding the demographic characteristics of respondents and (ii) questions regarding influencing factors and consumer intention to utilize renewable energy. The description of the data collected is provided in Table 1.

3.3.2. Measures and data collection

The study of Liu et al. (2013) was consulted, and its scale items were used to measure the perception of self-effectiveness. The

Table 1
Survey information.

Study parameters	Value
Time	May, June, and July (2019)
Location	Rawalpindi, Lahore, Gujranwala, and Faisalabad
Sample size	400
Valid responses	351
Response rate	87.70%

work of Kaldellis et al. (2012) was considered, and its scale items associated with environmental concern were extracted. The research of Moula (2013) was chosen to obtain the scale items linked with RE cost and RE benefits. The study of Alam et al. (2014) was consulted, and its scale items for evaluating RE awareness were used. The research of Kalkbrenner and Roosen (2016) was considered, and its scale items for measuring perception about neighbor participation were obtained. Finally, the work of Shakeel and Rahman (2018) was selected, and its scale items related to consumer intention were collected. A five-point Likert scale was used to scrutinize all items, with 1 indicating “strongly disagree” and 5 indicating “strongly agree”. An aggregated result of all scale items was then considered as the projected value of these variables.

3.3.3. Data analysis

Data analysis was conducted using SPSS (V. 26) and AMOS (V. 26). Hypotheses were tested by employing Structural Equation Modeling (SEM), which is a robust method for examining the associations among different variables and has three major advantages over conventional multivariate methods: (i) a clear estimation of measurement errors, (ii) calculation of latent (unobserved) constructs by observed constructs, and (iii) model testing for the implementation and evaluation of a structure in compliance with the data. Many of the multivariate methods implicitly ignore measurement error by not directly measuring it. However, SEM methods measure both independent and dependent variables with these error variance factors (Novikova et al., 2013). The method delivers precise and meaningful results due to its robustness and reliability (Hair et al., 2014). Consequently, we have utilized SEM because it is the most suitable approach for investigating the associations among all the variables used in the study.

4. Results

4.1. Sample characteristics

Table 2 represents the demographic profiles of participants, in which 57.3% of participants were male. The highest proportion of participants were middle-aged (44.4%), aged between 36 and 55 years, followed by young (31.9%) and old people (23.6%). A total of 37.3% of the participants had per month income between PKR 31,000 and 40,000 and were categorized as upper-middle-income-class, followed by middle-income-class (27.4%), with a per month income between PKR 21,000 and 30,000. Of these participants, 34.2% graduated from college, and 25.4% had a middle school education. Technical personnel (51.6%) constituted the largest part of the sample, followed by the “own business” group (20.8%).

4.2. Summary statistics and the test of discriminant validity

The summary statistics of the data are analyzed using the mean and standard deviation. Significant correlations are obtained among variables by using the Pearson correlation test. The issue of discriminant validity is tested by utilizing the square root of the average variance extracted (AVE) with the objective of determining the degree to which the elements of the variables are theoretically

¹ The data on the urban and total population of Pakistan have been obtained from the Worldometers website, available at <https://www.worldometers.info/world-population/pakistan-population/>.

Table 2
Demographic profiles of participants.

Features	Range	Frequency	(%)
Gender	Male	201	57.3
	Female	150	42.7
Age (years)	18–35	112	31.9
	36–55	156	44.4
	Above 55	83	23.6
Income (PKR)	< 10,000	19	5.4
	10,000–20,000	66	18.8
	21,000–30,000	96	27.4
	31,000–40,000	131	37.3
	41,000–50,000	27	7.7
	> 50,000	12	3.4
Education	Uneducated	16	4.6
	Primary school	66	18.8
	Middle school	89	25.4
	College degree	120	34.2
	Graduate degree	60	17.1
Occupation	Technical personnel	181	51.6
	Government servant	16	4.60
	Own business	73	20.8
	Farmer	58	16.5
	Others	23	6.6

Notes: Technical personnel are those people certified in the use and operation of electric power equipment.

not interconnected. It is confirmed that there is no discriminant validity problem because the square root of the AVE is more than its correlation to other variables (Irfan et al., 2020b) (see Table 3).

4.3. Validity and reliability

Cronbach's Alpha was calculated to check the internal reliability of items. Cronbach's Alpha values surpassed the minimum threshold value of 0.70 for each variable, as suggested by Nunnally (1978), indicating that the data are valid and reliable (see Table 4).

4.4. Content validity

Content validity reflects the adequacy of which an individual content domain has been tested, i.e., whether an instrument is truly an inclusive measure of the studied area. It is subjective and judgmental in its determination (Nunnally, 1978). Fig. 2 verifies the content validity of the model because each item was loaded on the corresponding structure.

4.5. Composite reliability and convergent validity test

The test of Composite Reliability (CR) was executed to evaluate the accuracy of each variable's items. Table 4 shows that the

CR values for all variables exceed the minimum accepted value, as recommended by Nunnally (1978). The test of Convergent Validity was conducted by using the AVE and item loadings to determine the degree to which the items are theoretically related (Wong, 2013). The outcomes show that the values of the AVE surpass 0.50, highlighting that the latent variables retained at least 50% variance.

4.6. Factor analysis

To determine the contribution structure, Exploratory Factor Analysis (EFA) was performed. The Kaiser-Meyer-Olkin (KMO) test and Bartlett's Test of Sphericity (BTS) were conducted to determine the sampling adequacy of the data. The results uncover that the KMO value is 0.919, specifying that data factor analysis can be performed (Kaiser, 1974). Moreover, the BTS value was also significant (9958.990), which satisfied the EFA requirements (see Table 5). Next, Confirmatory Factor Analysis (CFA) was carried out to check the data adequacy of the suggested model.

4.7. Hypothesis testing and estimation of the structural equation

The structural model was estimated, and the hypothesized relationships were analyzed after obtaining valid and reliable measures. As an essential step, the R^2 value was computed to assess the variations in the outcome variable explained by the explanatory variables. Cohen (2013) suggested that the R^2 value should not be less than 0.35. Our analysis shows that the R^2 value is 0.75, which surpasses the threshold value, suggesting considerable interpretation. To test all relationships proposed in the model, curve estimation and the SEM algorithm based on covariance were performed, which provided a high F -value, specifying that the model has linear relationships. A collinearity diagnostic test was performed to check for a multicollinearity problem. Field (2009) indicated that the value of the Variance Inflation Factor (VIF) should not exceed 10. The results show that the multicollinearity problem does not exist in the model since the VIF values of all variables did not exceed the recommended value.

The schematic representation of SEM is presented in Fig. 3. Table 6 reports the structural model's goodness-of-fit indices. It is obvious from Table 6 that the values of all fit indices are within the recommended criterion (Lucianetti et al., 2018), denoting that the data have been absolutely incorporated by the model. From the obtained results, the path coefficient value (β) of the relationship between the PSE and IURE is 0.70, with a p -value < 0.001, indicating that consumers' IURE is significantly and positively affected by their PSE. Thus, hypothesis 1 ($H1$) was accepted. It was found that BCRE have a negative effect on consumers' IURE, as the path coefficient value (β) of the relationship between BCRE and IURE is -0.00, with a p -value < 0.05. Therefore, we accepted hypothesis 2 ($H2$). Similarly, the path coefficient values (β) of the relationship between ARE and IURE and between PNP and IURE are 0.10 and

Table 3
Correlation matrix and discriminant validity findings.

Variables	M	SD	ECN	PNP	BBRE	ARE	BCRE	PSE	IURE
ECN	4.26	.755	[0.713]						
PNP	2.99	.946	0.329	[0.826]					
BBRE	3.79	1.22	0.274	0.485	[0.822]				
ARE	2.39	.937	0.354	0.375	0.523	[0.755]			
BCRE	2.41	1.10	0.172	0.543	0.413	0.303	[0.782]		
PSE	4.10	.594	0.346	0.248	0.171	0.329	0.220	[0.838]	
IURE	3.53	.441	0.295	0.568	0.507	0.418	0.723	0.233	[0.743]

Notes: The square root of the AVE is shown in brackets []; **M**: Mean; **SD**: Standard deviation; **IURE**: Intention to utilize renewable energy.

Table 4
Factor loadings along with the results of convergent validity.

Variables	Statements	Standard loadings	AVE	Composite reliability	Cronbach's alpha
Perception of self-effectiveness			0.702	0.922	0.919
PSE1	I have the knowledge necessary to utilize RE	0.731			
PSE2	I have complete control over utilizing RE	0.811			
PSE3	I have all RE utilization resources	0.905			
Environmental concern		0.509	0.805	0.805	
ECN1	I am concerned about pollution	0.726			
ECN2	I am concerned about environmental issues	0.740			
ECN3	I am concerned about climate crises	0.695			
ECN4	The environment could be improved by using RE	0.680			
Beliefs about the cost of RE utilization			0.611	0.887	0.893
BCRE1	The utilization of RE creates additional costs	0.869			
BCRE2	Renewables are costly, as a massive investment is required for RE projects	0.953			
BCRE3	The cost of RE consumption is high due to substantial installment costs	0.741			
Awareness of RE			0.569	0.902	0.904
ARE1	I have the awareness that the market is providing RE-based solutions	0.553			
ARE2	I have the awareness about the advantages of RE usage	0.829			
Perception about neighbor participation			0.683	0.937	0.939
PNP1	Neighbor participation encourages me to utilize RE	0.780			
PNP2	Neighbor participation motivates me to utilize RE	0.807			
Beliefs about the benefits of RE			0.676	0.936	0.937
BBRE1	Using RE decreases GHG emissions and improves energy composition	0.653			
BBRE2	The usage of RE would strengthen public scenery	0.843			
BBRE3	The usage of RE would improve energy supply	0.814			
BBRE4	New RE installations would create employment opportunities	0.865			
Intention to utilize RE			0.553	0.832	0.826
IURE1	I intend to utilize RE	0.687			
IURE2	This energy-saving attitude provokes me to utilize RE	0.753			
IURE3	I intend to pay more for RE than for other energy sources	0.646			
IURE4	I strongly suggest that other people use RE	0.581			

Notes: Method of rotation: Promax; Method of extraction: Maximum likelihood.

Table 5
Kaiser–Meyer–Olkin (KMO) and Bartlett's tests.

KMO and Bartlett's test		
Kaiser-Meyer-Olkin Measure of Sampling Adequacy		0.919
Bartlett's Test of Sphericity	Approx. Chi-square	9,958.990
	df	57
	Sig.	0.000

Notes: Cumulative variance explained: 63.97%; df: Degrees of freedom; Sig: Significance.

Table 6
Goodness-of-fit indices of the SEM model.

Fit statistics	Value	Recommended criterion	Description
X ² /df	1.362	< 3 good fit	Chi-square
CFI	0.990	> 0.9 good fit	Comparative fit index
IFI	0.991	> 0.9 good fit	Incremental fit index
GFI	0.981	> 0.9 good fit	Goodness-of-fit index
NFI	0.965	> 0.9 good fit	Normed fit index
TLI	0.979	> 0.9 good fit	Tucker-Lewis index
RMSEA	0.032	< 0.08 good fit	Root mean squared error of approximation
SRMR	0.02	< 0.09 good fit	Standardized root mean squared residual

0.09, with p -values < 0.01 and < 0.05, respectively. The results reveal that consumers' IURE is significantly affected by ARE and PNP. Thus, hypotheses 4 (H_4) and 5 (H_5) were accepted. In contrast, the path coefficients of the relationships between ECN and IURE and between BBRE and IURE failed to support hypotheses 2 (H_2) and 6 (H_6), respectively. As a result, ECN and BBRE do not have a significant effect on consumers' IURE, and therefore, hypotheses 2 (H_2) and 6 (H_6) are rejected. Table 7 reports the structural paths and hypothesis validity.

4.8. Additional testing

Endogeneity testing is mainly used to maintain the robustness of the research results (Jean et al., 2016). If there is endogeneity

Table 7
Structural paths and hypothesis results.

Hypotheses	Structural paths	Coef (β)	Conclusion	VIF	R ²
H1	PSE→IURE	0.70***	Accepted	1.289	0.75
H2	ECN→IURE	0.09	Rejected	1.389	
H3	BCRE→IURE	-0.00*	Accepted	1.850	
H4	ARE→IURE	0.10**	Accepted	1.659	
H5	PNP→IURE	0.09*	Accepted	1.873	
H6	BBRE→IURE	0.06	Rejected	1.830	

Notes: Significance level (*** p < 0.001, ** p < 0.01, and * p < 0.05).

in the data, it can possibly threaten the research findings. In addition, the maximum likelihood estimate could be distorted due to endogeneity bias, posing a significant threat to the validity of the

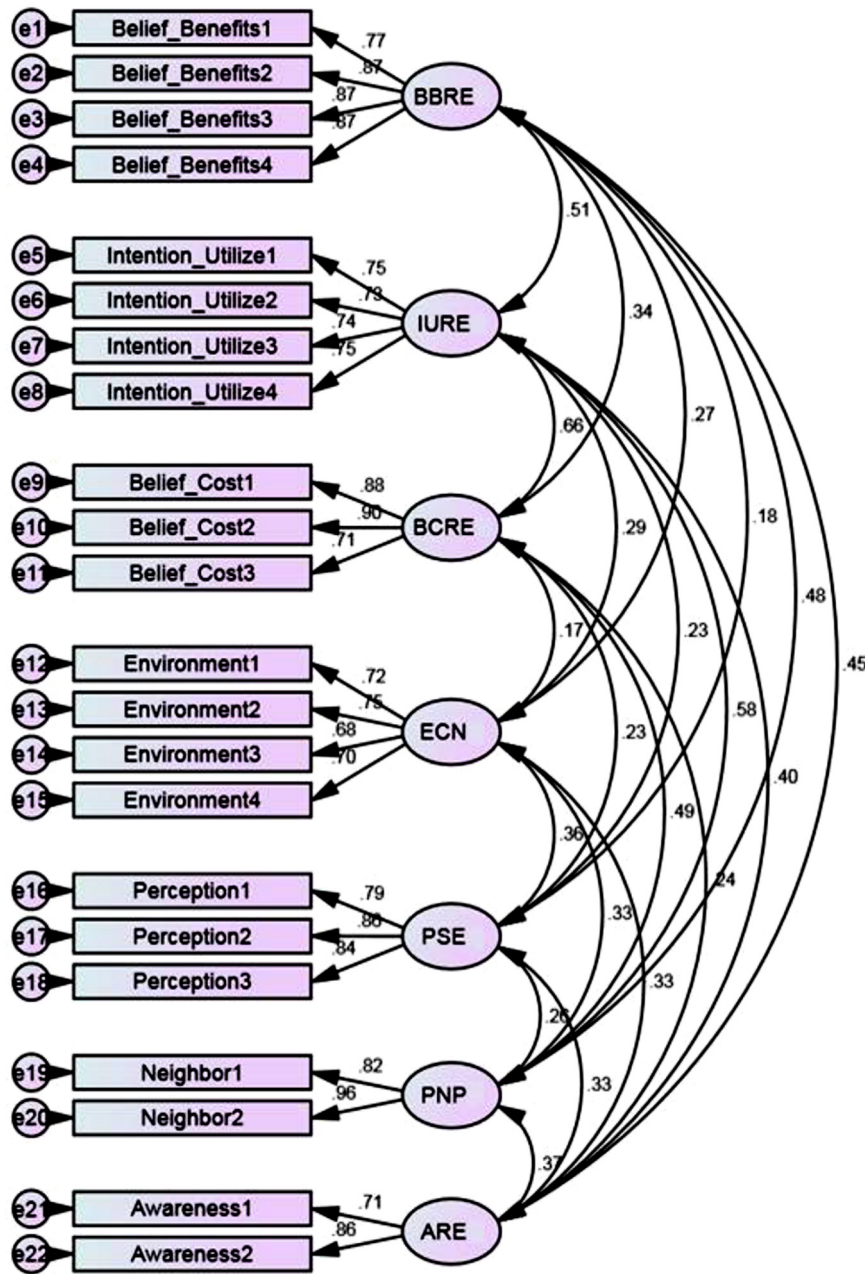


Fig. 2. Measurement model.

Table 8
Endogeneity test.

Hypotheses	Structural paths	Coef (β)	t-Statistics	Conclusion
H1	PSE → IURE	0.071***	3.059	Not different
H2	ECN → IURE	0.582	-0.288	Not different
H3	BCRE → IURE	-0.005***	17.956	Not different
H4	ARE → IURE	0.072**	2.383	Not different
H5	PNP → IURE	0.045*	1.863	Not different
H6	BBRE → IURE	0.080	2.778	Not different

Notes: Significance level (***p < 0.001, **p < 0.01, and *p < 0.05).

results. To overcome this, we performed the Heckman test by using the Stata program to obtain propensity values while examining endogeneity. The same significance was obtained as in the previous model, specifying that our results do not suffer from endogeneity bias (see Table 8).

5. Discussion

5.1. Relationship between PSE and IURE

The findings of our study demonstrate that the PSE has a positive and significant effect on consumers' IURE and are consistent with prior investigations (Wojuola and Alant, 2017; Zhang et al., 2011). Energy efficiency, energy security, and the eco-friendly character of RE could be the potential drivers that encourage residents to utilize RE. Out of various available energy generation sources, such as oil-fired power generators and urgent power supply (UPS) in Pakistan, the utilization of PV for domestic usage is more convenient and simple and has a longer life duration (Jabeen et al., 2019). Buyers' technology evaluation and understanding of its use would help develop confidence in RETs as significant dynamics in the coming years.

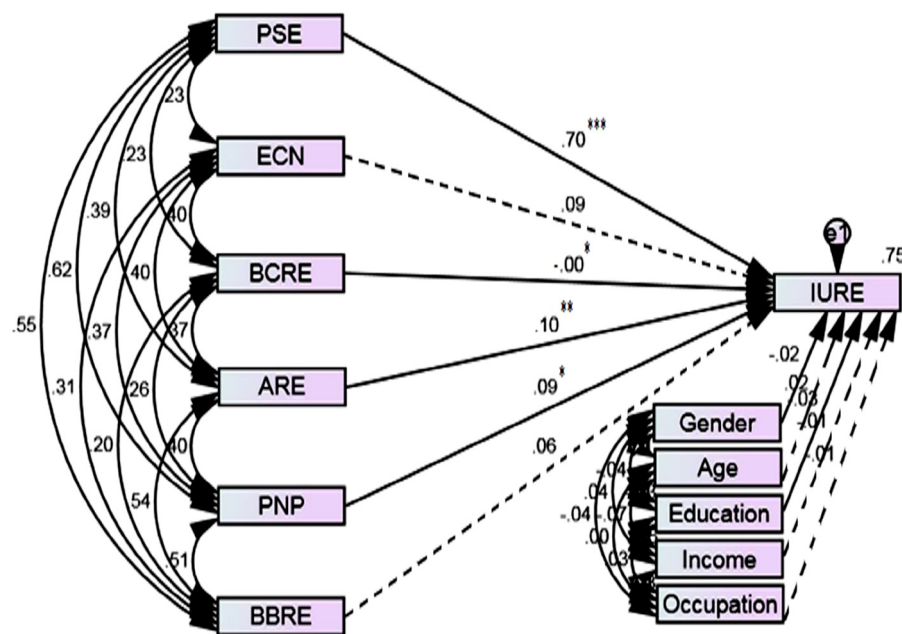


Fig. 3. Schematic representation of SEM. Smooth lines indicate significant paths, and broken lines indicate insignificant paths. ***, **, and * represent statistical significance at the 0.001, 0.01, and 0.05 levels, respectively.

5.2. Relationship between ECN and IURE

It is commonly understood that RETs help improve the environment and can reduce carbon production discharges. Prior studies have investigated how the resident intention to utilize RE is affected by ECN (Kalkbrenner and Roosen, 2016; Wang et al., 2018). Based on these findings, it was anticipated that similar outcomes may occur in Pakistan as well. The results of our study, however, show that ECN does not have a significant influence on consumers' IURE. The potential cause could be connected to the vital desire for which people buy such power generation solutions. In addition, unlike governments who value climate vulnerability problems, control carbon emissions, and expect technology to genuinely substitute for the reduction in energy prices, achieve long-term environmental benefits, and gradually contribute to environmental improvement, residents in Pakistan focus less on climate issues and pollution bottlenecks during their purchase decisions (Shakeel et al., 2018; Ikram et al., 2019). The absence of a strong policy mechanism is one of the primary reasons for this kind of behavior (Irfan et al., 2020b). The Pakistani government seldomly makes it easier for people to have awareness about environmental problems, their obligations, and the active role they can play in promoting the environment.

5.3. Relationship between BCRE and IURE

RE-based power solutions are expensive compared to traditional energy generation methods, and the higher purchasing cost has turned out to be a key barrier to their adoption. For instance, it currently costs approximately 5–6 times more to set up PV systems for domestic purposes than it does to set up other available energy generation techniques, such as UPS and power generators. A normal Pakistani household cannot bear such above-average charges and thus refrain from using RE solutions. The research results aid our supposition because cost has a negative impact on consumers' IURE. The findings of prior studies endorsed our results, as Sardonou and Genoudi (2013) and Iskin et al. (2013) reported that cost is a significant hurdle to RE adoption. Moreover, Komendantova and Yazdanpanah (2017) reported that this high

cost often influences the consumer intention and discourages RETs' acceptance.

5.4. Relationship between ARE and IURE

The results reveal that awareness has a positive and significant impact on consumers' IURE, specifying that residents with a good knowledge of RE are more likely to have a positive attitude toward it and tend to utilize it. These results are in accordance with the findings of a previous study (Ali et al., 2019). Similarly, Abdullah et al. (2017) found that awareness positively affects consumer intention to use RE. The possible reason is that residents have awareness of RE and the advantages it offers compared with conventional energy sources. Due to energy crises, the majority of residents are becoming aware that RE use may resolve these crises and ultimately fix the energy shortage problems. Consumer awareness about RE technology, markets, and support for installation is rapidly increasing, which will be useful in the upcoming years.

5.5. Relationship between the PNP and IURE

The previous findings uncover that the PNP has an encouraging effect on consumers' IURE. Our results are coherent with previous findings (Fischer and Sauter, 2004). Kosugi et al. (2019) reported that resident intention to adopt PV systems is positively affected by their PNP. In a developing country with scarce resources such as Pakistan, the social system is very integrated, and the actions of friends, family, neighbors, peers, and society have a significant effect on decision making (Shakeel and Rahman, 2018). The former knowledge of peers about using RE might affect the behavior of residents in a manner in which a pleasant experience enables RE acceptance, whereas an unpleasant experience has a negative impact on buyer intention. These findings signify that the PNP plays a significant role during RE purchase decisions.

5.6. Relationship between BBRE and IURE

The findings of the study reported that consumers' BBRE has no substantial effect on their intention to use RE. These outcomes

showed that residents' buying behaviors are based on their positive belief of the benefits about a particular product they want to purchase (Stigka, 2014). If residents are aware of the perceived benefits of RE use, then they would accept it. Yasmin and Grundmann (2019) reported that citizens may not have adequate knowledge about the benefits of RETs, which is the major barrier to RE adoption. However, during RE deployment in the country, the Pakistani government has not promoted the comprehensive benefits of RE, and citizens still favor conventional energy sources, which are inexpensive in comparison to RETs. Furthermore, residents' belief in RETs is limited; therefore, RE acceptance is at its preliminary phase in the country.

6. Conclusions

This study examines the impact of influencing factors on consumer intention to accept and use RE. Possible factors have been addressed in this analysis that may inspire or prevent people from using RE. By incorporating three novel and integral factors (the perception of self-effectiveness, beliefs about the benefits of RE, and perception about neighbor participation), this study has expanded the theoretical framework of the TPB to comprehensively analyze the factors that might shape public behavior regarding RE acceptance and utilization, which is the major contribution of this study. In the four large cities of Pakistan, an inclusive survey was arranged, and data analysis was performed using the SEM. The research results disclose interesting findings with significant policy proposals and decision-making insights. Consumers' IURE is positively and significantly influenced by their awareness, perception of self-effectiveness, and perception about neighbor participation. Conversely, it is negatively influenced by beliefs about the cost of RE utilization. On the other hand, beliefs about RE benefits and environmental concern do not have significant influences on consumer intention.

This study serves as a useful guide for the policy makers, organizations, and stakeholders involved in the RETs business. Based on the research findings, it is recommended that the importance of mitigating climate change, saving energy, and making eco-friendly energy supply choices urgently needs to be stressed. Public awareness of environmental problems and the benefits offered by RE to address these problems should be increased by all stakeholders, such as the government, NGOs, and public and private institutions. This can be accelerated by organizing climate and environment-related awareness-raising workshops and campaigns at the state level. The government should conduct awareness sessions on RE utilization and acceptance. Almost all RE development projects are capital-intensive and require high initial investment, i.e., solar energy parks, wind power plants, biomass plants, and hydropower projects. Therefore, the government should provide developers with simple and long-term financial support. A unique loan quota should be recommended to banks and other monetary organizations for RE development programs. To encourage national and multinational companies to invest in RE programs, the interest rate should be lowered. The findings show that the beliefs of citizens regarding RE costs have a negative impact on their IURE. Therefore, the Pakistani government should introduce subsidy schemes to encourage domestic consumers to use RE.

Pakistan has an integrated social system, denoting that there is a deep influence of the actions of friends, family, neighbors, and society on consumers' minds. Social standards should therefore be reshaped by enabling residents to choose RE over conventional energy sources. Public cooperation should be enhanced to authenticate that the national policy targets of integrating RE in country's energy portfolio are generating the desired outcomes. Enterprises dealing with RE development should substitute old marketing tactics with novel business concepts to fully commercialize RETs. A

robust policy mechanism is necessary to improve consumer intention and public acceptance of RE in the long run. The government must replace the old energy policies that are typically intended for traditional energy sources with new and transparent policies to quickly deploy RE in the country.

The off-grid RE system is an effective and appropriate technology. The system could benefit individuals, as the shutdown of electricity during the summer season lasts 8–10 hours in urban areas and 14–18 hours in rural areas in Pakistan. R&D is also an essential value-adding component of the renewable sector. At the local level, continuous R&D activities can improve production facilities, which in turn would minimize production costs and the reliance on foreign markets.

The current study also has some limitations that should be taken into account when conducting further research in this area. First, the authors did not measure attitude toward RE, which is an important factor in behavioral studies. It would produce better results if this factor was analyzed. Second, four large cities were considered for data collection purposes, and rural areas were not included. Throughout urban and rural areas, socioeconomic characteristics such as income, awareness, and education can differ considerably. Future researchers can tackle this limitation by taking inputs from rural dwellers in addition to urban dwellers. Finally, it could be interesting to analyze the relationship between awareness and beliefs about the benefits of RE, which the current study did not address; thus, future research could be conducted on this relationship.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table A.1
Factors influencing consumer intention to utilize RE.

Variables	Statements	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
Perception of self-effectiveness						
PSE1	I have the knowledge necessary to utilize RE					
PSE2	I have complete control over utilizing RE					
PSE3	I have all RE utilization resources					
Environmental concern						
ECN1	I am concerned about pollution					
ECN2	I am concerned about environmental issues					
ECN3	I am concerned about climate crises					
ECN4	The environment could be improved by using RE					
Beliefs about the cost of RE utilization						
BCRE1	The utilization of RE creates additional costs					
BCRE2	Renewables are costly, as a massive investment is required for RE projects					
BCRE3	The cost of RE consumption is high due to substantial installment costs					
Awareness of RE						
ARE1	I have awareness about the market providing RE-based solutions					
ARE2	I have awareness about the advantages of RE usage					
Perception about neighbor participation						
PNP1	Neighbor Participation encourages me to utilize RE					
PNP2	Neighbor participation motivates me to utilize RE					
Beliefs about the benefits of RE						
BBRE1	Using RE decreases GHG emissions and improves energy composition					
BBRE2	The usage of RE would strengthen public scenery					
BBRE3	The usage of RE would improve energy supply					
BBRE4	New RE installations would create employment opportunities					
Intention to utilize RE						
IURE1	I intend to utilize RE					
IURE2	Energy-saving attitudes provoke me to utilize RE					
IURE3	I intend to pay more for RE than for other energy sources					
IURE4	I strongly suggest that other people use RE					

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.spc.2020.10.031](https://doi.org/10.1016/j.spc.2020.10.031).

Appendix A. Survey questionnaire

Table A.1.

References

- Aitken, M., 2010. Wind power and community benefits: Challenges and opportunities. *Energy Policy* 38 (10), 6066–6075.
- Ajzen, I., 1991. The theory of planned behavior. *Organizational behavior and human decision processes* 50 (2), 179–211.
- Akhtar, J., Yaqub, M.I., Iqbal, J., Sheikh, N., Saba, T., 2018. Way forward in meeting energy challenges in Pakistan. *International Journal of Ambient Energy* 39 (8), 904–908.
- Alam, S.S., Hashim, N.H., Rashid, M., Omar, N.A., Ahsan, N., Ismail, M.D., 2014. Small-scale households renewable energy usage intention: Theoretical development and empirical settings. *Renew Energy* 68, 255–263.
- Ali, G., Yan, N., Hussain, J., Xu, L., Huang, Y., Xu, S., Cui, S., 2019. Quantitative assessment of energy conservation and renewable energy awareness among variant urban communities of Xiamen, China. *Renew Sustain Energy Rev* 109, 230–238.
- Al-Marri, W., Al-Habaibeh, A., Watkins, M., 2018. An investigation into domestic energy consumption behaviour and public awareness of renewable energy in Qatar. *Sustain Cities Soc* 41, 639–646.
- Awadh, A.I., Hassali, M.A., Al-Jela, O.Q., Bux, S.H., Elkalimi, R.M., Hadi, H., 2014. Immunization knowledge and practice among Malaysian parents: a questionnaire development and pilot-testing. *BMC Public Health* 14 (1), 1107.
- Azapagic, A., Stichnothe, H., 2011 Jan 1. Life cycle sustainability assessment of biofuels. In: *Handbook of biofuels production*. Woodhead Publishing, pp. 37–60.
- Batley, S.L., Colbourne, D., Fleming, P.D., Urwin, P., 2001. Citizen versus consumer: challenges in the UK green power market. *Energy Policy* 29 (6), 479–487.
- Batley, S.L., Fleming, P.D., Urwin, P., 2000. Willingness to pay for renewable energy: Implications for UK green tariff offerings. *Indoor Built Environ* 9 (3–4), 157–170.
- Čabelková, I., Strielkowski, W., Firsova, I., Korovushkina, M., 2020. Public Acceptance of Renewable Energy Sources: a Case Study from the Czech Republic. *Energies* 13 (7), 1742.
- Caspary, G., 2009. Gauging the future competitiveness of RE in Colombia. *Energy Econ* 31 (3), 443–449.
- Cohen, J., 2013. *Statistical power analysis for the behavioral sciences*. Academic press, New York, p. 490.
- Dalton, G.J., Lockington, D.A., Baldock, T.E., 2008. A survey of tourist attitudes to renewable energy supply in Australian hotel accommodation. *Renew Energy* 33 (10), 2174–2185.
- Devine-Wright, P., 2005. Beyond NIMBYism: towards an integrated framework for understanding public perceptions of wind energy. *Wind Energy: Int J Prog Appl Wind Power Convers Technol* 8 (2), 125–139.
- Düşteğör, D., Sultana, N., Felemban, N., Al-Qahtani, D., 2015. Public acceptance of RE and Smart-Grid in Saudi Arabia. 2015 IEEE 8th GCC Conference & Exhibition (pp. 1–6). IEEE.
- Etikan, I., Musa, S.A., Alkassim, R.S., 2016. Comparison of convenience sampling and purposive sampling. *American J Theor Appl Stat* 5 (1), 1–4.
- Field, A., 2009. *Discovering statistics using SPSS*. SAGE Publications Ltd, London 166–81.
- Fischer, C., Sauter, R., 2004. Users as pioneers: Transformation in the electricity system, microCHP and the role of the users. In: *Proceedings Berlin Conference on Human Dimensions of Global Environmental Change*, pp. 319–337.
- Gordon, N., 2010. Sustainable information technology awareness. *Innov Teach Learn Inform Comput Sci* 9 (2), 1–4.
- Hagen, B., Pijawka, D., 2015. Public perceptions and support of renewable energy in North America in the context of global climate change. *Int J Disaster Risk Sci* 6 (4), 385–398.
- Hair Jr, J.F., Sarstedt, M., Hopkins, L., Kuppelwieser, V.G., 2014. Partial least squares structural equation modeling (PLS-SEM). *Eur Bus Rev* 26, 106–121.
- Hamilton, L.C., Bell, E., Hartter, J., Salerno, J.D., 2018. A change in the wind? US public views on renewable energy and climate compared. *Energy Sustain Soc* 8 (1), 11.
- Hansla, A., Gamble, A., Juliusson, A., Gärling, T., 2008. Psychological determinants of attitude towards and willingness to pay for green electricity. *Energy policy* 36 (2), 768–774.
- Hartmann, P., Apaolaza-Ibáñez, V., 2012. Consumer attitude and purchase intention toward green energy brands: The roles of psychological benefits and environmental concern. *J Bus Res* 65 (9), 1254–1263.
- Hosseini, A., Zolfaghazadeh, M.M., Asghar Sadabadi, A., Aslani, A., Jafari, H., 2018. Social Acceptance of RE in Developing Countries: Challenges and Opportunities. *Distrib Gen Altern Energy* 33 (1), 31–48.
- Huijts, N.M., Molin, E.J., Steg, L., 2012. Psychological factors influencing sustainable energy technology acceptance: A review-based comprehensive framework. *Renew Sustain Energy Rev* 16 (1), 525–531.
- Ikram, M., Zhang, Q., Sroufe, R., Shah, S.Z., 2020. Towards a sustainable environment: the nexus between ISO 14001, renewable energy consumption, access to electricity, agriculture and CO2 emissions in SAARC countries. *Sustain Prod Consum* 22, 218–230.

- Ikram, M., Zhou, P., Shah, S.A., Liu, G.Q., 2019. Do environmental management systems help improve corporate sustainable development? Evidence from manufacturing companies in Pakistan. *J Clean Prod* 20 (226), 628–641.
- Iqbal, T., Dong, C.Q., Lu, Q., Ali, Z., Khan, I., Hussain, Z., Abbas, A., 2018. Sketching Pakistan's energy dynamics: Prospects of biomass energy. *J Renew Sustain Energy* 10 (2), 023101.
- Irfan, M., Zhao, Z.Y., Ahmad, M., Mukeshimana, M.C., 2019a. Critical factors influencing wind power industry: A diamond model based study of India. *Energy Rep* 5, 1222–1235.
- Irfan, M., Zhao, Z.Y., Ahmad, M., Mukeshimana, M.C., 2019b. Solar energy development in Pakistan: Barriers and policy recommendations. *Sustainability* 11 (4), 1206.
- Irfan, M., Zhao, Z.Y., Ahmad, M., Rehman, A., 2019c. A techno-economic analysis of off-grid solar PV system: a case study for Punjab province in Pakistan. *Processes* 7 (10), 708.
- Irfan, M., Zhao, Z.Y., Li, H., Rehman, A., 2020b. The influence of consumers' intention factors on willingness to pay for renewable energy: a structural equation modeling approach. *Environ Sci Pollut Res* 1–5.
- Irfan, M., Zhao, Z.Y., Mukeshimana, M.C., Ahmad, M., 2019d. 2019d. Wind Energy Development in South Asia: Status, Potential and Policies. In: 2019 2nd International Conference on Computing, Mathematics and Engineering Technologies (iCoMET), pp. 1–6 IEEE.
- Irfan, M., Zhao, Z.Y., Panjwani, M.K., Mangi, F.H., Li, H., Jan, A., Ahmad, M., Rehman, A., 2020a. Assessing the energy dynamics of Pakistan: Prospects of biomass energy. *Energy Rep* 6, 80–93.
- Irfan, M., Zhao, Z.Y., Rehman, A., Ozturk, I., Li, H., 2020c. Consumers' intention-based influence factors of renewable energy adoption in Pakistan: a structural equation modeling approach. *Environ Sci Pollut Res* 1–14. doi:10.1007/s11356-020-10504-w, In press.
- Iskin, I., Taha, R.A., Daim, T.U., 2013. Exploring the adoption of alternative energy technologies: a literature review. *Int J Sustain Soc* 5 (1), 43–61.
- Islam, T., 2014. Household level innovation diffusion model of photo-voltaic (PV) solar cells from stated preference data. *Energy Policy* 65, 340–350.
- Jabeen, G., Yan, Q., Ahmad, M., Fatima, N., Qamar, S., 2019. Consumers' intention-based influence factors of renewable power generation technology utilization: A structural equation modeling approach. *J Clean Prod* 237, 117737. doi:10.1016/j.jclepro.2019.117737, In press.
- Jean, R.J., Deng, Z., Kim, D., Yuan, X., 2016. Assessing endogeneity issues in international marketing research. *Int Mark Rev* 33 (3), 483–12.
- Käberger, T., 2018. Progress of renewable electricity replacing fossil fuels. *Glob Energy Interconnect* 1 (1), 48–52.
- Kaiser, H.F., 1974. An index of factorial simplicity. *Psychometrika* 39 (1), 31–36.
- Kaldellis, J.K., Kapsali, M., Katsanou, E., 2012. Renewable energy applications in Greece—What is the public attitude? *Energy Policy* 42, 37–48.
- Kalkbrenner, B.J., Roosen, J., 2016. Citizens' willingness to participate in local renewable energy projects: The role of community and trust in Germany. *Energy Res Soc Sci* 13, 60–70.
- Karlström, H., Ryghaug, M., 2014. Public attitudes towards renewable energy technologies in Norway. The role of party preferences. *Energy Policy* 67, 656–663.
- Kaufmann, H.R., Panni, M.F., Orphanidou, Y., 2012. Factors affecting consumers' green purchasing behavior: An integrated conceptual framework. *Amfiteatru Econ J* 14 (31), 50–69.
- Kaygusuz, K., 2012. Energy for sustainable development: A case of developing countries. *Renew Sustain Energy Rev* 16 (2), 1116–1126.
- Khambalkar, V.P., Katkhede, S.S., Dahatonde, S., Korpe, N.D., Nage, S.M., 2010. Renewable energy: an assessment of public awareness. *Int J Ambient Energy* 31 (3), 133–142.
- Kim, Y., Choi, S.M., 2005. Antecedents of green purchase behavior: An examination of collectivism, environmental concern, and PCE. *ACR North American Advanc* 592–99.
- Klepcka, A.M., Florkowski, W.J., Meng, T., 2018. Clean, accessible, and cost-saving: Reasons for rural household investment in solar panels in Poland. *Resour Conserv Recycl* 139, 338–350.
- Klepcka, A.M., 2019. Significance of renewable energy sources in sustainable development. *Roczniki (Annals)* 1230–2019–3873.
- Komendantova, N., Yazdanpanah, M., 2017. Impacts of human factors on willingness to use renewable energy sources in Iran and Morocco. *Environ Energy Econ Res* 1 (2), 141–152.
- Kostakis, I., Sardanou, E., 2012. Which factors affect the willingness of tourists to pay for renewable energy? *Renew Energy* 38 (1), 169–172.
- Kosugi, T., Shimoda, Y., Tashiro, T., 2019. Neighborhood influences on the diffusion of residential photovoltaic systems in Kyoto City, Japan. *Environ Econ Policy Stud* 21 (4), 477–505.
- Liu, W., Wang, C., Mol, A.P., 2013. Rural public acceptance of renewable energy deployment: The case of Shandong in China. *Appl Energy* 102, 1187–1196.
- Lopes, J.R.N., de Araújo Kalid, R., Rodríguez, J.L.M., Ávila Filho, S., 2019. 2019. A new model for assessing industrial worker behavior regarding energy saving considering the theory of planned behavior, norm activation model and human reliability. *Resour Conserv Recycl* 145, 268–278.
- Lucianetti, L., Jabbour, C.J., Gunasekaran, A., Latan, H., 2018. Contingency factors and complementary effects of adopting advanced manufacturing tools and managerial practices: Effects on organizational measurement systems and firms' performance. *Int J Prod Econ* 200, 318–328.
- Majhi, R., Banerjee, A., 2019. Journal of Environmental Accounting and Management. *J Environ Account Manag* 7 (3), 279–290.
- McEachan, R.R., Conner, M., Taylor, N.J., Lawton, R.J., 2011. Prospective prediction of health-related behaviours with the theory of planned behaviour: A meta-analysis. *Health Psychol Rev* 5 (2), 97–144.
- McGowan, F., Sauter, R., 2005. Public opinion on energy research: a desk study for the research councils. SPRU. University of Sussex, Sussex Energy Group.
- Mirza, B., 2015. Energy poverty and the perception of, and satisfaction with, renewable energy technologies: The case of solar villages in Pakistan. In: *Sustainable Access to Energy in the Global South*. Springer, Cham, pp. 113–127.
- Mohandes, N., Sanfilippo, A., Al Fakhri, M., 2019. Modeling residential adoption of solar energy in the Arabian Gulf Region. *Renew Energy* 131, 381–389.
- Mohsin, M., Abbas, Q., Zhang, J., Ikram, M., Iqbal, N., 2019. Integrated effect of energy consumption, economic development, and population growth on CO2 based environmental degradation: a case of transport sector. *Environ Sci Pollut Res* 26, 32824–32835. https://doi.org/10.1007/s11356-019-06372-8.
- Moula, M.M.E., Nyári, J., Bartel, A., 2013. Public acceptance of biofuels in the transport sector in Finland. *I J Sustain Built Environ* 6 (2), 434–441.
- Musall, F.D., Kuik, O., 2011. Local acceptance of renewable energy—A case study from southeast Germany. *Energy Policy* 39 (6), 3252–3260.
- Ndebele, T., 2020. Assessing the potential for consumer-driven renewable energy development in deregulated electricity markets dominated by renewables. *Energy Policy* 136, 111057.
- Novikova, S.I., Richman, D.M., Supekar, K., Barnard-Brak, L., Hall, D., 2013 Jan 1. NDAR: A model federal system for secondary analysis in developmental disabilities research. In: *International review of research in developmental disabilities*, 45. Academic Press, pp. 123–153.
- Nunnally, J.C., 1978. *Psychometric Theory*. McGraw-Hill, New York, NY.
- Opiyo, N.N., 2019. Impacts of neighbourhood influence on social acceptance of small solar home systems in rural western Kenya. *Energy Res Soc Sci* 52, 91–98.
- Osborne, J.W., Costello, A.B., 2004. Sample size and subject to item ratio in principal components analysis. *Pract Assess Res Eval* 9 (1), 11.
- Palm, A., 2017. Peer effects in residential solar photovoltaics adoption—A mixed methods study of Swedish users. *Energy Res Soc Sci* 26, 1–10.
- Paravantis, J.A., Stigka, E., Mihalakakou, G., Michalena, E., Hills, J.M., Dourmas, V., 2018. Social acceptance of renewable energy projects: A contingent valuation investigation in Western Greece. *Renew Energy* 123, 639–651.
- Parkinson, J., David, P., Rundle-Thiele, S., 2017. Self-efficacy or perceived behavioural control: Which influences consumers' physical activity and healthful eating behaviour maintenance? *J Consum Behav* 16 (5), 413–423.
- Procter, L., Angus, D.J., Blaszczyński, A., Gainsbury, S.M., 2019. Understanding use of consumer protection tools among Internet gambling customers: Utility of the Theory of Planned Behavior and Theory of Reasoned Action. *Addict Behav* 99, 106050.
- Qureshi, T.M., Ullah, K., Arentsen, M.J., 2017. Factors responsible for solar PV adoption at household level: A case of Lahore, Pakistan. *Renew Sustain Energy Rev* 78, 754–763.
- Rai, V., Reeves, D.C., Margolis, R., 2016. Overcoming barriers and uncertainties in the adoption of residential solar PV. *Renew Energy* 89, 498–505.
- Ram, M., Aghahosseini, A., Breyer, C., 2020. Job creation during the global energy transition towards 100% renewable power system by 2050. *Technol Forecast Soc Change* 151, 119682.
- Raza, M.Y., Wasim, M., Sarwar, M.S., 2020. Development of Renewable Energy Technologies in rural areas of Pakistan. *Energy Sourc Part A: Recover Util Environ Eff* 42 (6), 740–760.
- Rehman, E., Ikram, M., Feng, M.T., Rehman, S., 2020. Sectoral-based CO2 emissions of Pakistan: a novel Grey Relation Analysis (GRA) approach. *Environ Sci Pollut Res* 27, 29118–29129. https://doi.org/10.1007/s11356-020-09237-7.
- Sardanou, E., Genoudi, P., 2013. Which factors affect the willingness of consumers to adopt renewable energies? *Renew Energy* 57, 1–4.
- Shah, S.A., Valasai, G.D., Memon, A.A., Laghari, A.N., Jalbani, N.B., Strait, J.L., 2018. Techno-economic analysis of solar pv electricity supply to rural areas of Balochistan, Pakistan. *Energies* 11 (7), 1777.
- Shakeel, S.R., Rahman, S.U., 2018. Towards the establishment of renewable energy technologies' market: An assessment of public acceptance and use in Pakistan. *J Renew Sustain Energy* 10 (4), 045907.
- Sharma, N.K., Tiwari, P.K., Sood, Y.R., 2012. Solar energy in India: Strategies, policies, perspectives and future potential. *Renew Sustain Energy Rev* 16 (1), 933–941.
- Si, H., Shi, J.G., Tang, D., Wu, G., Lan, J., 2020. Understanding intention and behavior toward sustainable usage of bike sharing by extending the theory of planned behavior. *Resour Conserv Recycl* 152, 104513.
- Sniehotta, F.F., Presseau, J., Araújo-Soares, V., 2014. Time to retire the theory of planned behaviour. *Health Psychol Rev* 1–7.
- Steg, L., Perlaviciute, G., van der Werf, E., 2015. Understanding the human dimensions of a sustainable energy transition. *Front Psychol* 6, 805.
- Stigka, E.K., Paravantis, J.A., Mihalakakou, G.K., 2014. Social acceptance of renewable energy sources: A review of contingent valuation applications. *Renew Sustain Energy Rev* 32, 100–106.
- Süsser, D., Kannen, A., 2017. 'Renewables? Yes, please!': perceptions and assessment of community transition induced by renewable-energy projects in North Frisia. *Sustain Sci* 12 (4), 563–578.
- Vand, B., Hast, A., Bozorg, S., Li, Z., Syri, S., Deng, S., 2019. Consumers' attitudes to support green energy: a case study in Shanghai. *Energies* 12 (12), 2379.
- Wang, B., Wang, Q., Wei, Y.M., Li, Z.P., 2018. Role of renewable energy in China's energy security and climate change mitigation: An index decomposition analysis. *Renew Sustain Energy Rev* 90, 187–194.

- Wang, Y, Xu, L, Solangi, YA, 2020. Strategic renewable energy resources selection for Pakistan: Based on SWOT-Fuzzy AHP approach. *Sustain Cities Soc* 52, 101861.
- Wojuola, RN, Alant, BP., 2017. Public perceptions about renewable energy technologies in Nigeria. *African J Sci Technol Innov Dev* 9 (4), 399–409.
- Wong, KK, 2013. Partial least squares structural equation modeling (PLS-SEM) techniques using SmartPLS. *Mark Bull* 24 (1), 1–32.
- Wüstenhagen, R, Wolsink, M, Bürer, MJ, 2007. Social acceptance of renewable energy innovation: An introduction to the concept. *Energy Policy* 35 (5), 2683–2691.
- Yasmin, N, Grundmann, P., 2019. Adoption and diffusion of renewable energy–The case of biogas as alternative fuel for cooking in Pakistan. *Renew Sustain Energy Rev* 101, 255–264.
- Yue, B, Sheng, G, She, S, Xu, J, 2020. Impact of Consumer Environmental Responsibility on Green Consumption Behavior in China: The Role of Environmental Concern and Price Sensitivity. *Sustainability* 12 (5), 2074.
- Yun, S, Lee, J., 2015. Advancing societal readiness toward renewable energy system adoption with a socio-technical perspective. *Technol Forecast Soc Change* 95, 170–181.
- Zhang, Y, Yu, Y, Li, T, Zou, B, 2011. Analyzing Chinese consumers' perception for biofuels implementation: The private vehicles owner's investigating in Nanjing. *Renew Sustain Energy Rev* 15 (5) 2299–09.
- Zografakis, N, Sifaki, E, Pagalou, M, Nikitaki, G, Psarakis, V, Tsagarakis, KP, 2010. Assessment of public acceptance and willingness to pay for renewable energy sources in Crete. *Renew Sustain Energy Rev* 14 (3), 1088–1095.